

feature which already formed part of the state of the art could not, by definition, make a contribution over the prior art and did not, therefore, qualify as a unifying element within the meaning of R. 13.1 PCT. According to R. 13.2 PCT, as in force from 1.7.1992, an international application could relate to a group of inventions if there was a "technical relationship" among those inventions involving one or more of the same or corresponding "special technical features", i.e. such technical features that defined a contribution which each of the claimed inventions made over the prior art. R. 13.1 PCT did not simply require some link between a group of inventions claimed in an international application, but a common inventive concept. This meant that there must be either a common technical problem or at least, if there was more than one technical problem, there must be one single technical concept behind the solutions of these different problems.

In T 957/96 the application described a number of processes which did not have any process step in common. According to the board, the decisive fact was that all the processes claimed shared a common technical feature, namely the use of the substantially pure regioisomer, which was essential for solving the technical problem addressed by the application. It was this use of the intermediate compound which formed the common "inventive" concept of all the claimed process variants. This feature constituted a special technical feature which defined the contribution that the claimed invention made over the prior art as required by R. 30(1) EPC 1973 (see now R. 44(1) EPC).

In W 11/99 (OJ 2000, 186) the board held that if, in an international application, there are claims directed to products and to a process for the manufacture thereof, it could not be assumed that there were no corresponding special technical features within the meaning of R. 13.2 PCT simply because the process could also be used to manufacture other products. The board interpreted the requirement for the presence of "corresponding special technical features" under R. 13.2 PCT in cases where a manufacturing process and products were being claimed in the same application to mean that they could usually be assumed to be present where the production process is new and actually suited to making the claimed products accessible (where appropriate in addition to further products). In such cases the board regards the process as being "specially adapted" to the manufacture of the claimed products. A narrower interpretation of the terms "specially adapted" and "corresponding special technical features" would not fulfil the legislative purpose of Art. 34(3) PCT and associated R. 13.1 PCT, which, in the board's view, was the same as that of Art. 82 EPC 1973, namely to prevent subject-matters which were not interconnected being claimed in one and the same application. This interpretation accorded with Annex B to the PCT Administrative Instructions mentioned in the PCT Guideline III-7.2 (July 1998 version), where it is stated in Part 1, item (e), that a process is specially adapted for the manufacture of a product if it inherently results in the product (see now Annex B, para. (e) PCT Administrative Instructions, as in force from 1.1.2022). If this condition was met, it was irrelevant whether other products could be obtained using the process.

In T 106/06 the board referred to the established case law of the boards of appeal according to which a manufacturing process and its resulting products are considered as unitary subject matter (e.g. W 2/95 and W 11/99). The board gave consideration to the question whether the fact that claim 1 did not relate to a manufacturing process in the

usual sense which may be characterised in that the specific end products are envisaged at the outset of the process, but to a process of isolating genes which, in view of its set up, has resemblance to a screening process which, in turn, may be characterised in that the final product is not known at the outset, should have an influence on the assessment of unity. The board held that the decisive question was whether the product had actually been produced by the process and not whether it had been known at the start of it. Therefore, there was no difference in the assessment of unity between a manufacturing process and a screening process and their resulting products.

In W 18/01 the board pointed out that it followed from the definitions in the Regulations under the PCT and the PCT International Preliminary Examination Guidelines (which are binding on the EPO; see G 1/89, OJ 1991, 155, point 6 of the Reasons) that it was normally not sufficient merely to define and examine the common core of a group of inventions, for example by indicating the features which were specified in all the claims defining the inventions of the group. R. 13.2 PCT required an examination of the technical relationship among the inventions in a group. Such a relationship could exist even if the inventions did not involve the same technical features, provided they involved corresponding special technical features. The definition of "special technical features" given in R. 13.2 PCT required an analysis of the contribution which each of the claimed inventions made over the prior art. This first entailed analysing which of the features distinguished the claimed inventions from the cited prior art before their contribution could be examined in the light of the description, in particular the problems solved and the effects achieved by the claimed inventions.

5.3. Unity of single claims defining alternatives ("Markush claims")

R. 44(1) EPC and R. 13.2 PCT also apply where a single claim defines alternatives ("Markush claim"). According to R. 44(2) EPC and R. 13.3 PCT, the determination of whether a group of inventions is so closely linked as to form a single general inventive concept shall be made without regard to whether the inventions are claimed in separate claims, or as alternatives within a single claim (see also W 35/91).

In considering whether the ISA had substantiated its findings of non-unity, the board in W 3/94 (OJ 1995, 775) referred to the section on the "Markush practice" in Annex B, Part 1(f) of the Administrative Instructions under the PCT, as in force from 1.7.1992 (see now Annex B, para. (f) PCT Administrative Instructions, as in force from 1.1.2019). Here it is stated that the requirement of a technical interrelationship and the same or corresponding special technical features as defined in R. 13.2 PCT shall be considered met when the alternatives are of a similar nature. The section then specifies when alternatives are to be regarded as "of a similar nature."

In W 1/94, the EPO acting as ISA had found that the compounds according to the "inventions" it had identified did not have a new structural element. Nevertheless the board concluded that the absence of such an element did not automatically destroy the unity of the invention. On the contrary, it was clear from the PCT Administrative Instructions that a technical relationship must be recognised for a group of alternative chemical compounds, such as all the compounds having a common property or activity, and such a common

activity existed in the case in point. The new use could therefore form the general inventive concept linking the subject-matters of the claims in accordance with R. 13.1 PCT and had to be treated as a functional characteristic linking the compounds claimed.

In W 6/95 the board referred to the PCT Administrative Instructions, binding not only on the ISA but also on the board of appeal acting as the "three-member board" according to R. 40.2(c) PCT (see G 1/89, OJ 1991, 155; since 1.4.2005, the references in the PCT Regulations to "three-member board" have been replaced by "review body" – see also in this chapter II.B.1. "Introduction"). The board noted that it followed from Annex B, Part 1(f)(i) PCT Administrative Instructions (now part of Annex B, para. (f) PCT Administrative Instructions, as in force from 1.1.2022) that in order to establish unity of invention "a posteriori", it was not sufficient for all alternatives of chemical compounds covered by a Markush claim to have a common property or activity, i.e. be suitable for solving a common technical problem, because according to item (B)(1) they had, in addition, to have a "significant structural element" in common in order for the alternatives to form unity. The board made it clear that it was, however, not in agreement with the explanation given in Annex B, Part 1(f)(ii) PCT Administrative Instructions to assume that the said "significant structural element" had to be novel per se. Rather, this expression meant that in relation to the said common property or activity there had to be a common part of the chemical structure which distinguished the claimed compounds from known compounds having the same property or activity (see also W 6/97).

In T 169/96 the board noted that the fact that claim 1 also comprised a known compound, not covered by claim 2, was of no relevance to the question of unity, because R. 30(b) EPC 1973 (in the version as in force until 31.5.91; also in the version in force thereafter) did not require that there must be a common concept unifying different "means" according to it. It was not relevant here that the three different classes of chemical compounds identified by the examining division comprised completely different chemical structures of a residue. The board observed that Chapter C-III, 7.4a of the then applicable EPO Guidelines, which had been relied on by the examining division, as well as Annex B, Part 1(f) PCT Administrative Instructions (see now Annex B, para. (f) PCT Administrative Instructions as in force from 1.1.2022), concerning Markush-type claims, rightly stated that the said significant structural element might consist of a combination of individual components linked together. It was not stated there that the combination of individual structural elements had to be novel per se, nor did such a requirement follow from Art. 82 EPC 1973. Rather it followed from that guideline that this expression meant that in relation to the said common property or activity there had to be a common part of the chemical structure which distinguished the claimed compounds from known compounds having the same property or activity (see now Guidelines F-V, 3.2.5 – November 2022 version).

In W 4/96 (OJ 1997, 552) the board noted that the requirement of a technical relationship as defined in R. 13.2, first sentence, PCT might be met when all claimed alternatives belong to a class of compounds which might be expected to behave in the same way in the context of the claimed inventions ("Markush claims"). The technical relationship involved those common special technical features that defined a contribution over the state of the art (R. 13.2, second sentence, PCT; see also W 6/96, W 1/10). However, such

contribution could not be recognised on the basis of this expectation if members of the class had already been shown in the prior art to behave in the manner disclosed in the application. The board added that if at least one Markush alternative was not novel over the prior art, the question of unity had to be reconsidered by the examiner, i.e. a posteriori (Annex B, Part I (f)(v) PCT Administrative Instructions in conjunction with PCT Search Guidelines; see now Annex B PCT Administrative Instructions, as in force from 1.1.2022, also PCT International Search and Preliminary Examination Guidelines, as in force from 1.3.2022).

The board also held that an objection of lack of unity could be raised a posteriori with regard to all categories of grouping of alternatives of chemical compounds. An a posteriori assessment of unity had to be made in all circumstances in the same way, since the legal requirement of R. 13 PCT was the same for all cases. Therefore, the a posteriori assessment of unity in cases of a functional relationship within a group of compounds had to be made in a manner corresponding to the assessment in cases of a structural relationship. This was appropriate for compounds related by their structure as well as for compounds related by their function. The structural relationship within the meaning of Annex B, Part I (f)(i)(B)(1) and the relationship within the meaning of Annex B, Part I (f)(i)(B)(2) and (iii) served the same purpose: both tests were intended to show whether the alternative compounds were of a similar nature, as was said in the introductory part of Annex B, Part I (f)(i). These tests were only tools in order to determine whether there was unity of invention as defined in R. 13 PCT.

Claim 1 in W 34/07 was directed to compositions for the coating of ceramic supports comprising micronised glass frits **and/or** nanosised zirconium hydroxide. The board concluded that the two claimed alternatives neither had in common express features not known in combination from the prior art nor represented two solutions to the same technical problem. The effects achieved by using either micronised glass or nanosised zirconium hydroxide were different and unrelated, although they might lead to "best results", i.e. when using a composition comprising both components.

6. Plurality of inventions – further search fees

If the search division or the ISA raises an objection of lack of unity, a further search fee is payable under R. 64(1) EPC (R. 46(1) EPC 1973) or Art. 17(3)(a) PCT, respectively, for each additional invention to be covered by the second report (for Euro-PCT applications in the European phase, see in this chapter II.B.6.2.).

In W 2/10 the board pointed out that in W 4/85 (OJ 1987, 63) and many subsequent decisions, the boards had decided that the requirement to give reasons in an invitation under Art. 17(3)(a) PCT was so fundamental that an unsubstantiated invitation could be regarded as legally ineffective. This was so in the case in question and therefore the additional search fees had to be refunded. See also OJ 2017, A20 for the provisional opinion now issued together with the partial search report.

If the fee is paid in due time applicants may select the single invention or group of inventions which they wish to retain in the (main) application. According to T 158/12, with

reference to G 2/92, this choice cannot be altered once examination of that invention has commenced. T 736/14 held that if an applicant responds unclearly or in a misleading way to an invitation from the examining division to specify which searched invention it wishes to prosecute further, then the division has to clarify this matter, e.g. via a further communication.

6.1. Consequences of non-payment of a further search fee

In view of differing decisions T 178/84 (OJ 1989, 157) and T 87/88 (OJ 1993, 430), the President of the EPO referred the following point of law to the Enlarged Board of Appeal:

"Can an applicant who fails to pay further search fees for a non-unitary application when requested to do so by the search division under R 46(1) EPC 1973 pursue that application for the subject-matter in respect of which no search fees have been paid or must he file a divisional application for it?"

In the opinion of the Enlarged Board, given in G 2/92 (OJ 1993, 591), the application could not be pursued for subject-matter in respect of which the applicants had not paid the further search fees. Instead the applicants had to file a divisional application for that subject-matter if they wished to continue to protect it. In the view of the Enlarged Board, it was clear from the procedural system of the EPC that the invention which was to be examined for patentability had to be an invention in respect of which a search fee had been paid prior to the drawing up of the European search report. Part IV of the EPC envisaged that an application progressed after filing from the search division to the examining division. One object of R 46 EPC 1973 was to implement this procedure by ensuring that an appropriately extensive search was completed in respect of each individual application before it was examined by the examining division. To this end, in response to an invitation from the search division to pay one or more further search fees in respect of one or more further inventions to which the application related, applicants had to pay such fees if they wished to ensure that one of the further inventions could become the subject of the claims of that application. That was the proper interpretation of R 46(1) EPC 1973 in context. This confirmed the practice according to T 178/84.

In T 319/96 the initial application lacked unity, but no further search fee was paid. The applicant had argued that, because of the international-type search (Art. 15(5)(a) PCT) carried out by the EPO (in the priority application) for the subject-matter of original claims 4 to 10, it had not forfeited its right to choose and could therefore continue pursuing this subject-matter. It had paid a search fee for each of the two inventions, and two search reports had been prepared by the Office. The board, however, stated that under R 46(1) EPC 1973 a search fee was to be paid for each further invention if the **European** search report was to cover that invention. R 46 EPC 1973 did not provide that a search report from another procedure could be substituted for the European search report. The board referred rather to the possibility under Art. 10(2) RFees (see now Art. 9(2) RFees) of a full or partial refund of the European search fee where the Office has prepared an earlier search report on the same subject-matter. G 2/92 applied also in the present case because in the context of the European procedure only one search report had been drawn up after payment of only one search fee.